

$$21. y = e^x [a \cos x + b \sin x - \cos x \log(\sin x + \tan x)],$$

$$22. y = a \cos x + b \sin x - 1 + \sin x - x \cos x + \sin x \log(1 + \sin x),$$

$$23. y = \frac{a}{x} + \frac{b}{x^2} + \frac{1}{3}e^x,$$

$$24. y = Ae^x + Be^{-x} + \frac{x(x-1)e^x}{4},$$

$$25. y = a + be^{2x} - \frac{1}{2}e^x \sin x,$$

$$26. y = ae^x + bxe^x - xe^x \sin x - 2e^x \cos x,$$

$$27. y = ax + \frac{b}{x+1} + \frac{x^2}{6} \frac{4x+3}{x+1},$$

$$28. y = ax + bxe^x - x(x+1),$$

$$29. y = ax^3 + \frac{b}{x} - \frac{1}{3}x^2 \left(\log x + \frac{2}{3} \right),$$

$$30. y = ax + \frac{b}{x} + e^x \left(1 - \frac{1}{x} \right),$$

$$31. y = a + bx^2 + (x-1)e^x,$$

$$32. y = ax + bxe^{2x} - \frac{x}{4} - \frac{x^2}{2},$$

$$33. y = a + bx + \frac{c}{x} - \log x.$$

6.7 Method of Undetermined Coefficients

The method of undetermined coefficients is also applied to find a particular solution of a differential equation. Though this method is simpler and economical than the method of variation of parameter, this method cannot be applied in a few cases which are discussed later on. We consider the problem of solving an equation of the form

$$F(D)y = [D^n + P_1 D^{n-1} + P_2 D^{n-2} + \cdots + P_{n-1} D + P_n]y = Q, \quad (i)$$

where $D = \frac{d}{dx}$.

In this method we assume the particular integral as

$$y_p = A_1 q_1(x) + A_2 q_2(x) + \cdots + A_s q_s(x), \quad (ii)$$

where $q_1(x), q_2(x), \dots, q_s(x)$ are the terms of Q and those arising from these terms by differentiation, for example, if the equation is $F(D)y = \sin x$, we take for (ii),

$$y_p = A \cos x + B \sin x,$$

if the equation is $F(D)y = x^2$, we take for (ii),

$$y_p = Ax^2 + Bx + C,$$

if the equation is $F(D)y = e^{2x} + e^{4x}$, we take for (ii),

$$y_p = Ae^{2x} + Be^{4x},$$

since no new term is obtained by differentiating e^{2x} , e^{4x} , and A_i 's are undetermined constants to be so chosen that (ii) satisfies (i). If Q is $\sec x$, $\operatorname{cosec} x$, $\tan x$ or $\cot x$, the method fails, since the number of terms obtained by differentiation of Q is infinite.

If a term of Q , say L , is also a term of the complementary function corresponding to an s -fold root m , then in (ii) we add a term $x^s L$ and the terms arising from it by differentiation. For example, for the equation $(D-1)^2(D+2)y = e^x + \sin x$, we take the particular integral as

$$y_p = Ax^2e^x + Bxe^x + Ce^x + D\sin x + E\cos x.$$

If a term of Q is $x^s L$ and L is also a term of the complementary function corresponding to an s -fold root m , then in (ii) we add a term $x^{s+s}L$ and the terms arising from it by differentiation. For example, for the equation $(D-1)^2(D+2)y = xe^x + x^3$, we take the particular integral as

$$y_p = Ax^3e^x + Bx^2e^x + Cxe^x + De^x + Ex^3 + Fx^2 + Gx + H.$$

When Q is an algebraic function then the particular integral of (6.7.1) will be of the form of (ii), if all P_i are non-zero. If $P_n = 0$, then we take the particular integral as

$$y_p = x[A_1q_1(x) + A_2q_2(x) + \dots + A_sq_s(x)].$$

If $P_n = 0$ and $P_{n-1} = 0$, then we take the particular integral as

$$y_p = x^2[A_1q_1(x) + A_2q_2(x) + \dots + A_sq_s(x)] \text{ and so on.}$$

WORKING RULE

We know that the general solution of the differential equation

$$y_2 + P(x)y_1 + Q(x)y = R(x)$$

has the form

$$y = \text{C.F.} + \text{P.I.}$$

The method of undetermined coefficients leads the way of finding the P.I. in the

following ways.

// **R1.** If $R(x) = e^{\lambda x}$ and λ is a multiple root of the auxiliary equation of order k , then we take the P.I. as

$$y_p = Ax^k e^{\lambda x},$$

where A is the undetermined constant. Obviously, $k = 0$, if λ is not a root of the auxiliary equation.

// **R2.** If $R(x) = a_0 + a_1x + a_2x^2 + \dots + a_kx^k$, i.e., a polynomial in x and $P(x)$, $Q(x)$ do not vanish, we take the P.I. as

$$y_p = A_0 + A_1x + A_2x^2 + \dots + A_kx^k.$$

When $Q(x) = 0$, we take

$$y_p = x(A_0 + A_1x + A_2x^2 + \dots + A_kx^k)$$

and if $P(x) = 0 = Q(x)$, then we take

$$y_p = x^2(A_0 + A_1x + A_2x^2 + \dots + A_kx^k).$$

// **R3.** If $R(x) = \sin ax$ or $\cos ax$ then we take

$$y_p = A\sin ax + B\cos ax,$$

if $\sin ax$ or $\cos ax$ is absent in the complementary function.

If $\sin ax$ or $\cos ax$ is present in the complementary function then we take

$$y_p = x(A \sin ax + B \cos ax).$$

R4. If $R(x) = e^{\lambda x} \sin bx$ or $e^{\lambda x} (a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k)$ or $\sin bx (a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k)$, then the P.I. will be modified according to the above mentioned rules.

For example, if $R(x) = e^{\lambda x} \sin ax$ and λ is not a root of the auxiliary equation, then we take the P.I. as

$$R(x) = e^{\lambda x} (A \sin ax + B \cos ax).$$

If $R(x) = e^{\lambda x} (a_0 + a_1 x + \dots + a_k x^k)$ and λ is a multiple root of the auxiliary equation of order k , then we take

$$y_p = x^k e^{\lambda x} (A_0 + A_1 x + \dots + A_k x^k).$$

If $R(x) = \sin bx (a_0 + a_1 x + \dots + a_k x^k)$ and $\sin bx$ is present in the complementary function, then we take

$$y_p = x(A \cos bx + B \sin bx) (A_0 + A_1 x + \dots + A_k x^k).$$

WORKED-OUT EXAMPLE

Solve the following equation by the method of undetermined coefficients:

$$y_2 + 2y_1 + 2y = e^{2x}.$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 + 2m + 2 = 0.$$

Solving, we get $m = -1 \pm i$. So the C.F. is

$$e^{-x} (A \cos x + B \sin x).$$

Since 2 is not a root of the auxiliary equation, we take the P.I. as

$$y_p = C e^{2x}.$$

Using this in the given equation, we get $C = \frac{1}{10}$. Hence the general solution of the given equation is

$$y = e^{-x} (A \cos x + B \sin x) + \frac{e^{2x}}{10}.$$

Solve the following equation by the method of undetermined coefficients:

$$y_2 + 4y_1 + 4y = 16e^{-2x}$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 + 4m + 4 = 0.$$

Solving, we get $m = -2, -2$. So the C.F. is

$$(c_1 + c_2 x) e^{-2x}.$$

Since -2 is a double root of the auxiliary equation, we take the P.I. as

$$y_p = Cx^2e^{-2x}.$$

Differentiating this, we get

$$y_p' = 2Cx(1-x)e^{-2x} \text{ and } y_p'' = -4Cx(1-x)e^{-2x} + 2C(1-2x)e^{-2x}.$$

Using the above results in the given equation, we get $C = 8$. Hence the general solution of the given equation is

$$y = (c_1 + c_2x)e^{-2x} + 8x^2e^{-2x}.$$

3. Solve the following equation by the method of undetermined coefficients:

$$y_2 + 4y = \sin x.$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 + 4 = 0.$$

Solving, we get $m = \pm 2i$. So the C.F. is

$$c_1 \cos 2x + c_2 \sin 2x.$$

Since, $\sin x$ is not a part of the complementary function, we take the P.I. as

$$y_p = A \cos x + B \sin x.$$

Using it in the given equation, we get

$$3A \cos x + 3B \sin x = \sin x.$$

This gives $B = \frac{1}{3}$, $A = 0$. Hence the general solution of the given equation is

$$y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{3} \sin x.$$

4. Solve the following equation by the method of undetermined coefficients:

$$y_2 + y = 2 \cos x.$$

Solution

Here the C.F. is $C \cos x + D \sin x$. Now we see that $\cos x$ is a part of Q and also occur in C.F. corresponding to a root of multiplicity 1. So, we assume the particular integral as

$$y_p = x(A \cos x + B \sin x).$$

Putting these values in the given equation and comparing both sides, we get $A = 0$, $B = 1$.

Hence, the general solution is

$$y = C \cos x + D \sin x + x \sin x.$$

5. Solve the following equation by the method of undetermined coefficients:

$$y_2 - 4y_1 + y = x^2 - 2x + 2.$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 - 4m + 1 = 0.$$

Solving, we get $m = 2 \pm \sqrt{3}$. So the C.F. is

$$e^{2x} [De^{\sqrt{3}x} + Ee^{-\sqrt{3}x}].$$

We take the P.I. as

$$y_p = Ax^2 + Bx + C,$$

which gives

$$(y_p)_1 = 2Ax + B \text{ and } (y_p)_2 = 2A.$$

Putting these values in the given equation and comparing both sides, we get $A = 1$, $B = 6$, $C = 24$. So the general solution is

$$y = e^{2x} [De^{\sqrt{3}x} + Ee^{-\sqrt{3}x}] + x^2 + 6x + 24.$$

6. Solve the following equation by the method of undetermined constants:

$$y_2 - 2y_1 = e^x \cos x.$$

Solution

Here the C.F. is $C + De^{2x}$. We take the particular integral as

$$y_p = Ae^x \cos x + Be^x \sin x.$$

This gives

$$(y_p)_1 = [A + B]e^x \cos x + [B - A]e^x \sin x \text{ and } (y_p)_2 = 2[B \cos x - A \sin x]e^x.$$

Putting these values in the given equation and comparing both sides, we get $A = -\frac{1}{2}$ and $B = 0$. So the general solution is

$$y = C + De^{2x} - \frac{1}{2}e^x \cos x.$$

7. Solve the following equation by the method of undetermined coefficients:

$$y_2 - y_1 - 2y = x^2 e^x.$$

Solution

Here the C.F. is $De^{-x} + Ee^{2x}$. We assume the particular integral as

$$y_p = [Ax^2 + Bx + C]e^x,$$

which gives

$$(y_p)_1 = [Ax^2 + (2A + B)x + B + C]e^x \text{ and } (y_p)_2 = [Ax^2 + (4A + B)x + 2A + 2B + C]e^x$$

Putting these values in the given equation and comparing both sides we get

$$A = -\frac{1}{2}, B = -\frac{1}{2} \text{ and } C = -\frac{3}{4}.$$

Hence, the general solution is

$$y = De^{-x} + Ee^{2x} - \frac{1}{4}[2x^2 + 2x + 3]e^x.$$

8. Solve the following equation by the method of undetermined coefficients:

$$y_2 - 4y_1 + 4y = x^3 e^{2x}.$$

Solution

Here the C.F. is $[E + xF]e^{2x}$. Now we see that e^{2x} is a part of Q and also occurs in C.F. corresponding to a root of multiplicity 2. So, we assume the particular integral as

$$y_p = [Ax^5 + Bx^4 + Cx^3 + Dx^2]e^{2x},$$

leaving the terms xe^{2x} , e^{2x} as those appear in the C.F. Putting these values in the given equation and comparing both sides, we get $A = \frac{1}{20}$, $B = C = D = 0$. Hence the general solution is

$$y = [E + xF]e^{2x} + \frac{1}{20}e^{2x}x^5.$$

9. Solve by the method of undetermined coefficients:

$$y_3 + 3y_2 + 2y_1 = x^2 + 4x + 8.$$

Solution

Here the auxiliary equation is

$$m^3 + 3m^2 + 2m = 0,$$

whose solutions are $m = 0, -1, -2$. So the C.F. is

$$C_1 + C_2e^{-x} + C_3e^{-2x}.$$

Here $P_3 = 0$, so we take the particular integral

$$y_p = x(A + Bx + Cx^2).$$

$$\text{Now } y'_p = A + 2Bx + 3Cx^2; y''_p = 2B + 6Cx; y'''_p = 6C.$$

Using these values in $y'''_p + 3y''_p + 2y'_p = x^2 + 4x + 8$, we get

$$6C + 3(2B + 6Cx) + 2(A + 2Bx + 3Cx^2) = x^2 + 4x + 8$$

Comparing both sides, we get

$$6C = 8; 4B + 18C = 4 \text{ and } 2A + 6B + 6C = 8.$$

Solving, we get

$$A = \frac{11}{4}; B = \frac{1}{4}; C = \frac{1}{6}.$$

So the general solution is

$$y = C_1 + C_2e^{-x} + C_3e^{-2x} + x\left(\frac{11}{4} + \frac{x}{4} + \frac{x^2}{6}\right).$$

10. Solve by the method of undetermined coefficients: $y_2 = x$.

Solution

Here the auxiliary equation is $m^2 = 0$,

whose solutions are $m = 0, 0$. So the C.F. is $C_1 + C_2x$.

We note here $P_2 = 0$ and $P_3 = 0$, so we take the particular integral

$$y_p = x^2(A + Bx).$$

Differentiating with respect to x , we get

$$y'_p = 2Ax + 3Bx^2 \text{ and } y''_p = 2A + 6Bx.$$

So,

$$y''_p = 2A + 6Bx = x \text{ gives } A = 0, B = \frac{1}{6}.$$

Hence, the general solution is

$$y = C_1 + C_2x + \frac{x^3}{6}.$$

1. Solve by the method of undetermined coefficients: $y_2 - 7y_1 + 6y = (x - 2)e^x$.

Solution
Here the C.F. is $Ee^x + Fe^{6x}$. Now we see that e^x is a part of Q and also occurs in C.F. corresponding to a root of multiplicity 1. So, we assume the particular integral as

$$y_p = (A + Bx)xe^x.$$

Differentiating with respect to x , we get

$$y'_p = (A + Ax + 2Bx + Bx^2)e^x \text{ and } y''_p = (2A + 2B + Ax + 4Bx + Bx^2)e^x.$$

Putting these values in $y''_p - 7y'_p + 6y_p = (x - 2)e^x$ and comparing both sides, we get $-10B = 1$ and $5A - 2B = 2$.

Solving we get $A = \frac{9}{25}$, $B = -\frac{1}{10}$. So

$$y_p = \left(\frac{9}{25} - \frac{1}{10}x\right)xe^x.$$

Hence, the general solution is

$$y = Ee^x + Fe^{6x} + \left(\frac{9}{25} - \frac{1}{10}x\right)xe^x.$$

Solve by the method of undetermined coefficients: $y_2 - 4y_1 + 4y = (x^2 + 1)xe^{2x}$.

Solution

Here the C.F. is $(F + xG)e^{2x}$. Now we see that e^{2x} is a part of Q and also occurs in C.F. corresponding to a root of multiplicity 2. So, we assume the particular integral as

$$y_p = (Ax^3 + Bx^2 + Cx + E)x^2e^{2x}.$$

Differentiating with respect to x , we get

$$y'_p = [2Ax^5 + (5A + 2B)x^4 + (4B + 2C)x^3 + (3C + 2E)x^2 + 2Ex]e^{2x} \text{ and}$$

$$y''_p = [4Ax^5 + 4(5A + B)x^4 + 4(5A + 4B + C)x^3 + 4(3B + 3C + E)x^2 + (6C + 8E)x + 2E]e^{2x}.$$

Putting these values in $y''_p - 4y'_p + 4y_p = (x^2 + 1)xe^{2x}$ and comparing both sides,

we get

$$20A = 1, 12B - 5E = 0, 6C = 1, E = 0.$$

Solving we get $A = \frac{1}{20}, B = 0, C = \frac{1}{6}, E = 0$.

So,

$$y_p = \left(\frac{1}{20}x^3 + \frac{1}{6}x \right) x^2 e^{2x}.$$

Hence, the general solution is

$$y = (F + xG)e^{2x} + \left(\frac{1}{20}x^3 + \frac{1}{6}x \right) x^2 e^{2x}.$$

13. Solve by the method of undetermined coefficients: $y_2 - 3y_1 = x + e^x \sin x$.

Solution

Here the C.F. is $A + Be^{3x}$. Since $P_3 = 0$ and Q has an algebraic part, we take the particular integral as

$$y_p = x(E + xF) + e^x(G \sin x + H \cos x).$$

Differentiating with respect to x , we get

$$y'_p = E + 2Fx + e^x(G - H) \sin x + e^x(G + H) \cos x,$$

and

$$y''_p = 2F + 2Ge^x \cos x - 2He^x \sin x.$$

Putting these values in $y''_p - 3y'_p = x + e^x \sin x$ and comparing both sides, we get

$$2F - 3E = 0, -6F = 1, H - 3G = 1, G + 3H = 0.$$

Solving, we get

$$E = -\frac{1}{9}, F = -\frac{1}{6}, G = -\frac{3}{10}, H = \frac{1}{10}.$$

So,

$$y_p = x \left(-\frac{1}{9} - \frac{x}{6} \right) - \frac{e^x}{10} (3 \sin x - \cos x).$$

14. Solve by the method of undetermined coefficients:

$$y_2 + y_1 - 6y = 10e^{2x} - 18e^{3x} - 6x - 11.$$

Solution

Here the C.F. is $Ae^{2x} + Be^{-3x}$. Since e^{2x} is a part of Q and also occur in C.F. corresponding to a root of multiplicity 1 we assume the particular integral as

$$y_p = Cxe^{2x} + Ee^{3x} + Fx + G.$$

Differentiating it with respect to x , we get

$$y'_p = (2x + 1)Ce^{2x} + 3Ee^{3x} + F, \text{ and } y''_p = 4(x + 1)Ce^{2x} + 9Ee^{3x}.$$

Using these values in $y''_p + y'_p - 6y_p = 10e^{2x} - 18e^{3x} - 6x - 11$, we get

$$5Ce^{2x} + 6Ee^{3x} - 6Fx + F - 6G = 10e^{2x} - 18e^{3x} - 6x - 11.$$

Comparing, we get

$$C = 2, E = -3, F = 1, G = 2.$$

So,

$$y_p = 2xe^{2x} - 3e^{3x} + x + 2.$$

Hence, the general solution is

$$y = Ae^{2x} + Be^{-3x} + 2xe^{2x} - 3e^{3x} + x + 2.$$

EXERCISE

• Solve the following equations by the method of undetermined coefficients:

1. $y_2 + 2y_1 + 2y = 8e^{2x}$,
2. $y_2 + 2y = e^x + 2$,
3. $y_2 - y_1 - 6y = 20e^{-2x}$,
4. $y_2 + y = 2\sin x$,
5. $y_2 - 4y_1 = x^2$,
6. $y_2 - 3y_1 = 2x^2 + 1$,
7. $y_2 - 9y = x + e^{2x} - \sin 2x$,
8. $y_2 - y = e^x \sin 2x$
9. $y_2 + 4y = \sin 2x$
10. $y_2 - 3y_1 + 2y = \sin 2x$,
11. $y_2 + 6y_1 + 9y = e^{-3x}$,
12. $y_2 - 2y_1 + 3y = x^3 + \sin x$,
13. $y_3 + 2y_2 - y_1 - 2y = x^2 + e^x$,
14. $y_3 - y_1 = x$,
15. $y_2 - 2y_1 + y = xe^x$,
16. $y_2 = x^2$,
17. $y_2 - 2y_1 + 5y = 25x^2 + 12$,
18. $y_2 + 4y = x^2 \sin 2x$,
19. $y_2 - 2y_1 = e^x \sin x$,
20. $y_2 + 3y_1 + 2y = e^x \sin x$,
21. $y_2 - 4y_1 + 3y = 12x^2 + x - 6$, given $y = 6$; $\frac{dy}{dx} = 11$, when $x = 0$.
22. $y_2 + 4y_1 = 12x^2 + e^x$, given that $y = 1$; $\frac{dy}{dx} = 1$, when $x = 0$.
23. $y_2 + y = 10e^{2x}$, given that $y = 0$; $\frac{dy}{dx} = 0$, when $x = 0$.
24. $y_2 + y = \cos 2x$, given that $y = 0$; $\frac{dy}{dx} = 1$, when $x = 0$.
25. $y_2 + 4y_1 + 5y = 5$, given that $y = 0$; $\frac{dy}{dx} = 0$, when $x = 0$.

ANSWER

1. $e^{-x}[A \cos x + B \sin x] + \frac{4}{5}e^{2x}$,
2. $A \cos \sqrt{2}x + B \sin \sqrt{2}x + 1 + \frac{1}{3}e^x$,
3. $Ae^{3x} + Be^{-2x} - 4xe^{-2x}$,
4. $A \cos x + B \sin x - x \cos x$,

5. $A + Be^{4x} - \frac{x^3}{12} - \frac{x^2}{16} - \frac{x}{32},$
6. $A + Be^{3x} - \frac{x}{27}(6x^2 + 6x + 13),$
7. $Ae^{3x} + Be^{-3x} - \frac{x}{9} - \frac{e^{2x}}{5} + \frac{1}{13}\sin 2x,$
8. $Ae^x + Be^{-x} - \frac{1}{8}[\sin 2x + \cos 2x]e^x,$
9. $A\cos 2x + B\sin 2x - \frac{x}{4}\cos 2x,$
10. $Ae^x + Be^{2x} - \frac{1}{20}\sin 2x + \frac{3}{20}\cos 2x,$
11. $(A + xB)e^{-3x} + \frac{1}{2}x^2e^{-3x},$
12. $e^x[A\cos\sqrt{2}x + B\sin\sqrt{2}x] + \frac{2x}{9} + \frac{2x^2}{3} + \frac{x^3}{3} - \frac{8}{27} + \frac{1}{4}(\sin x + \cos x),$
13. $Ae^x + Be^{-x} + Ce^{-2x} - \frac{x^2}{2} + \frac{x}{2} - \frac{5}{4} + \frac{xe^x}{6},$
14. $A + Be^x + Ce^{-x} - \frac{1}{2}x^2,$
15. $e^x[A + Bx] + \frac{1}{6}e^xx^3,$
16. $A + Bx + \frac{x^4}{12},$
17. $e^x(A\cos 2x + B\sin 2x) + 5x^2 + 4x + 2,$
18. $A\cos 2x + B\sin 2x - \frac{1}{12}x^3\cos 2x + \frac{1}{16}x^2\sin 2x + \frac{1}{32}x\cos 2x,$
19. $A + Be^{2x} - \frac{1}{2}e^x\sin x,$
20. $Ae^{-x} + Be^{-2x} + \frac{e^x}{10}(\sin x - \cos x),$
21. $-6e^x + 2e^{2x} + 4x^2 + 11x + 10,$
22. $\frac{29}{32} + \frac{3x}{8} - \frac{3x^2}{4} + x^3 + \frac{e^x}{5} - \frac{17}{160}e^{-4x},$
23. $2e^{2x} - 2\cos x - 4\sin x,$
24. $\sin x + \frac{1}{3}(\cos x - \cos 2x),$
25. $1 - e^{-2x}\cos x - 2e^{-2x}\sin x.$